A Knotted Universe: a new notion of reflective set theories

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Abstract. We describe a universe of set theories enjoying atoms without enduring infinite risk.

In the early 2000's Pitts and Gabbay's use of Fraenkl-Mostowski set theory (FM set theory, or just FM for short), a theory of sets with atoms, to model nominal phenomena, sparked a vibrant line of research. [7] [6]. As Chaitin points out, the axioms and assumptions of a theory comprise its risk [4]. This observation raises an interesting point: is it possible to have a set theory with atoms without taking on *infinite* risk? It turns out it is.

We describe a theory of sets that enjoys both the cleanliness of Zermelo-Fraenkl set theory (ZF set theory, or just ZF for short), that is arising only from the empty set, and yet with an infinite supply of atoms.

The shape of the solution is easy to state in advance: we take a familiar theory, make two copies of it, and tie them in a recursive knot, each copy serving as the supply of atoms for the other. When the author first arrived at this idea in 2006, copying a theory and binding the copies by mutual recursion was an unusual move, justified mostly by where it led. Two decades on it has become ordinary. In the age of AI and the autoformalization of mathematics and logic, copies of a theory are everywhere: autoformalization is exactly the business of producing a formal copy of an informal theory and reasoning about the fit between the two, proof assistants host reflective towers in which a theory manipulates a copy of itself, and a single development now routinely keeps several presentations of the same theory in play at once. Tying such copies into a recursive knot — a theory whose objects are those of its mirror image, and conversely — is no less natural: it is reflection[1], the self-referential idiom that runs from the reflection principles of logic through reflective calculi [9] to present-day systems that reason about their own output. So although the theory below was devised in 2006, it arguably reads more naturally in 2026 — the moves from which it is built have become part of the ordinary vocabulary of formalized mathematics.

1 Intuitions

We imagine two copies of FM set theory, say a red one and a black one. The red theory has red comprehensions, $\{\ldots\}$ and a red element-of relation, $\in$, while the black theory,

[1] We use *reflective* and *reflection* throughout in the self-referential sense of the reflective higher-order calculus — a theory that holds a copy of itself — and not in the sense of the Lévy–Montague reflection principles of set theory.

correspondingly has black comprehensions, $\{\ldots\}$ and a black element-of relation, $\in$. The red element-of relation, $\in$, can "see into" red sets, $\{\ldots\}$, but not black ones. Meanwhile, the black element-of relation, $\in$, can "see into" black sets $\{\ldots\}$, but not red ones. Thus, the black sets can serve as atoms for the red sets, while the red sets can serve as atoms for the black sets.

So, the two theories are mutually recursively defined. This mutual recursion is closely linked to game semantics in the sense of Abramksy, Hyland, and Ong, et al. Without loss of generality, we can think of red as player and black as opponent. A well-formed red set, S, is a winning strategy for player, while a well-formed black set, S, is a winning strategy for opponent.

2 Formal theory

The judgment $\Sigma; \Gamma \vdash S$ is pronounced "Σ thinks that S is well-formed, given dependencies, Γ ." Similarly, $\Sigma; \Gamma \vdash x$ is pronounced "Σ thinks that x is an admissible atom, given dependencies, Γ."

We think of X as some infinite supply of "atoms" and $\mathsf{S}[X]$ as a *stream* of atoms drawn from X. We assume basic stream operations on $\mathsf{S}[X]$, such as $\mathsf{take}(\mathsf{S}[X], n)$ which returns the pair (σ, Σ) where $\sigma = x_1 : \ldots : x_n$ and $\Sigma = \mathsf{drop}(\mathsf{S}[X], n)$. Given $\rho = \{j_1/i_1 : \ldots : j_n/i_n\}$ the operation $\mathsf{swap}(\mathsf{S}[X], \rho)$ denotes the application of the permutation ρ to $\mathsf{S}[X]$.

Thus, $\mathsf{S}[X]$ may be thought of as an ordering on X.

A rule of the form

$$\frac{H_1, \ldots, H_n}{C} R$$

is pronounced "R concludes that C given H_1, …, H_n".

We use $e, f, \ldots$ to range over sets and atoms.

2.1 Embracing content

The rules for judging when an atom is admissible or set is well formed are as follows

$$\frac{}{\Sigma; () \vdash \emptyset} \mathit{Foundation}$$

$$\frac{(x, \Sigma') = \mathsf{take}(\Sigma, 1) \quad \Sigma; \Gamma \vdash S}{\Sigma'; x, \Gamma \vdash \{x\} \cup S} \mathit{Atomicity}$$

$$\frac{\Sigma; \Gamma \vdash S}{\Sigma; \Gamma \vdash \{S\}} \mathit{Nesting}$$

$$\frac{\Sigma; \Gamma_1 \vdash S_1 \quad \Sigma; \Gamma_2 \vdash S_2}{\Sigma; \Gamma_1, \Gamma_2 \vdash S_1 \cup S_2} \mathit{Union}$$

$$\frac{\Sigma; \Gamma_1 \vdash S_1 \quad \Sigma; \Gamma_2 \vdash S_2}{\Sigma; \Gamma_1, \Gamma_2 \vdash S_1 \cap S_2} Intersection$$

$$\frac{\Sigma; \Gamma_1 \vdash S_1 \quad \Sigma; \Gamma_2 \vdash S_2}{\Sigma; \Gamma_1, \Gamma_2 \vdash S_1 \setminus S_2} Subtraction$$

2.2 Recognizing elements

$$\frac{\Sigma; \Gamma \vdash e}{\Sigma; \Gamma \vdash e \in \{e\}} Presence$$

$$\frac{\Sigma; \Gamma \vdash e}{\Sigma; \Gamma \vdash e \notin \emptyset} Emptiness$$

$$\frac{\Sigma; \Gamma \vdash S_1 \quad \Sigma; \Gamma \vdash S_2 \quad \Sigma; \Gamma \vdash e \in S_1}{\Sigma; \Gamma \vdash e \in S_1 \cup S_2} Inclusion$$

$$\frac{\Sigma; \Gamma \vdash S_1 \quad \Sigma; \Gamma \vdash S_2 \quad \Sigma; \Gamma \vdash e \in S_1 \quad \Sigma; \Gamma \vdash e \in S_2}{\Sigma; \Gamma \vdash e \in S_1 \cap S_2} Collusion$$

$$\frac{\Sigma; \Gamma \vdash S_1 \quad \Sigma; \Gamma \vdash S_2 \quad \Sigma; \Gamma \vdash e \in S_1 \quad \Sigma; \Gamma \vdash e \notin S_2}{\Sigma; \Gamma \vdash e \in S_1 \setminus S_2} Exclusion$$

2.3 Equations

The syntactic theory is too fine grained. It makes syntactic distinctions that do not correspond to distinct sets. We erase these syntactic distinctions with a set of equations on set expressions.

$$\frac{\Sigma; \Gamma \vdash e}{\Sigma; \Gamma \vdash e = e} Identity$$

$$\frac{\Sigma; \Gamma_1 \vdash_1 \;\; \Sigma; \Gamma_2 \vdash e_2 \; \Sigma; \Gamma_3 \vdash e_1 = e_2}{\Sigma; \Gamma_3 \vdash e_1 = e_2} Symmetry$$

$$\frac{\Sigma; \Gamma_1 \vdash e_1 \; \Sigma; \Gamma_2 \vdash e_2 \; \Sigma; \Gamma_3 \vdash e_3 \; \Sigma; \Gamma_4 \vdash e_1 = e_2 \; \Sigma; \Gamma_5 \vdash e_2 = e_3}{\Sigma; \Gamma_4, \Gamma_5 \vdash e_1 = e_3} Transitivity$$

$$\frac{\Sigma; \Gamma \vdash S}{\Sigma; \Gamma \vdash S \cup S = S} Idempotence_{\cup}$$

$$\frac{\Sigma; \Gamma \vdash S}{\Sigma; \Gamma \vdash S \cap S = S} Idempotence_{\cap}$$

2.4 Tying the first recursive knot

$$\frac{\Sigma; \Gamma \vdash e_1 \quad \Sigma; \Gamma \vdash e_2 \quad \Sigma; \Gamma \vdash e_1 \neq e_2}{\Sigma; \Gamma \vdash e_2 \notin \{e_1\}} Absence$$

$$\frac{\Sigma; \Gamma \vdash S \quad \Sigma; \Gamma_1 \vdash S_1 \quad \ldots \quad \Sigma; \Gamma_n \vdash S_2 \; \Sigma; \Delta \vdash f : S_1 \times \ldots \times S_n \to S}{\Sigma; \Gamma, \Gamma_1, \ldots, \Gamma_n, \Delta \vdash \{f(s_1, \ldots, s_n) \; : \; s_1 \in S_1, \; \ldots \; , s_n \in S_n\}} Comprehension$$

2.5 Set's signature

The signature of the theory is given by

$$\emptyset, |\emptyset| = 0 \qquad \mathsf{embrace}_i, |\mathsf{embrace}_i| = i \qquad \cap, |\cap| = 2 \qquad \cup, |\cup| = 2 \qquad \setminus, |\setminus| = 2$$

Which we write as $\mathsf{Set}[X]$.

3 Tying the second recursive knot

Now for the main course. We take two copies of the theory, that is two versions of the signature and the theories generated by each version. For simplicity, we write $\color{red}{\mathsf{Set}[X]}$ and $\mathsf{Set}[X]$. Then we set $\color{red}{X}= \mathsf{Set}[X]$ and $X =\color{red}{\mathsf{Set}[X]}$.

4 Fraenkel–Mostowski set theory, in brief

So that this paper stands on its own we recall the essentials of Fraenkel–Mostowski set theory in the form made popular by Gabbay and Pitts [7]; a reader already fluent in nominal techniques may skip directly to Section 5.

Atoms and the cumulative hierarchy. Fix a countably infinite set $\mathbb{A}$ whose elements we call *atoms*. Atoms are *not* sets: they carry no members of their own, yet they are permitted to occur as members of sets. Over $\mathbb{A}$ one erects the cumulative hierarchy of ZFA — Zermelo–Fraenkel set theory *with atoms* — by transfinite recursion,

$$\mathcal{U}_0 = \emptyset, \qquad \mathcal{U}_{\alpha+1} = \mathbb{A} \cup \mathcal{P}(\mathcal{U}_\alpha), \qquad \mathcal{U}_\lambda = \bigcup_{\alpha<\lambda} \mathcal{U}_\alpha \ \ (\lambda \text{ a limit}),$$

and sets $\mathcal{U} = \bigcup_{\alpha\in\mathrm{Ord}} \mathcal{U}_\alpha$. The axioms of ZFA are those of ZF, with extensionality, separation and replacement relaxed to leave the membersless atoms untouched.

The permutation action. Let $\mathrm{Sym}(\mathbb{A})$ be the group of all permutations (bijections) $\pi\colon \mathbb{A} \to \mathbb{A}$. Each π acts on the entire hierarchy by $\in$-recursion,

$$\pi \cdot a = \pi(a) \quad (a \in \mathbb{A}), \qquad \pi \cdot x = \{\, \pi \cdot y \ : \ y \in x \,\} \quad (x \text{ a set}),$$

and this is a group action: the identity acts trivially and $(\pi\sigma)\cdot x = \pi\cdot(\sigma\cdot x)$. Intuitively π renames atoms uniformly, reaching arbitrarily deep into a set but never *inside* an atom, which has no interior to reach.

Support and finite support. A set $w \subseteq \mathbb{A}$ *supports* x when every permutation that fixes w pointwise also fixes x:

$$\big(\forall a \in w.\ \pi(a) = a\big) \implies \pi \cdot x = x.$$

We call x *finitely supported* if some *finite* w supports it. The element is *equivariant* when the empty set supports it, i.e. when $\pi \cdot x = x$ for every π; equivariant data are precisely those nameable without reference to any particular atom.

The theory. Fraenkel–Mostowski set theory (FM) is ZFA augmented by the *finite-support axiom*: *every* element of the cumulative hierarchy is hereditarily finitely supported. Equivalently, one passes to the *Fraenkel–Mostowski universe*

$$\mathcal{U}_{\mathsf{FM}} \;=\; \{\, x \in \mathcal{U} \;:\; x \text{ is hereditarily finitely supported} \,\},$$

the largest subuniverse on which the $\mathrm{Sym}(\mathbb{A})$-action has only finitely-supported elements. Cutting the hierarchy down in this way is exactly the move that distinguishes FM from bare ZFA.

Least support, freshness, and the fresh quantifier. Two facts give FM its working calculus. First, for the symmetric group on atoms the intersection of two supports is again a support, so every finitely-supported x possesses a *least* support $\mathrm{supp}(x)$, a canonical finite set of atoms "x depends on". Second, $\mathbb{A}$ is infinite, so beyond any finite list of parameters there is always a fresh atom. One writes

$$a \# x \;\;:\!\iff\;\; a \notin \mathrm{supp}(x) \qquad (\text{“}a \text{ is fresh for } x\text{”}),$$

and introduces the *fresh quantifier*

$$\text{И}a.\, \phi(a) \;\;:\!\iff\;\; \{\, a \in \mathbb{A} : \neg\phi(a) \,\} \text{ is finite},$$

read "for a fresh atom a, $\phi(a)$". Its defining property is the *some/any theorem*: for any finitely-supported ϕ with parameters $\bar{p}$,

$$\big(\text{И}a.\, \phi(a)\big) \iff \big(\exists a \# \bar{p}.\; \phi(a)\big) \iff \big(\forall a \# \bar{p}.\; \phi(a)\big).$$

That "for some fresh atom" and "for all fresh atoms" coincide is the technical heart of the nominal approach to binding, and it is everything in this list — atoms, the permutation action, finite support, freshness — that a model of FM must reproduce. We turn to building one.

5 A model of FM in the reflective universe

We can now collect the dividend of the construction *Tying the second recursive knot* above. The reflective theory was assembled from the empty set alone, yet — we claim — it already contains a model of FM, and in fact two of them, interlocked. The atoms such a model needs are not postulated; they are *supplied by the dual copy.* Throughout we fix the two tied theories introduced above, the red copy $\color{red}{\mathrm{Set}[X]}$ and the black copy $\mathrm{Set}[X]$, with

$$\color{red}{X} \;=\; \mathrm{Set}[X], \qquad X \;=\; \color{red}{\mathrm{Set}[X]},$$

so that the admissible atoms of the red theory are exactly the black sets, and *vice versa.* We write the red copy in red and the black copy in black throughout, so that a red comprehension $\color{red}{\{\,\cdots\,\}}$ and a black comprehension $\{\,\cdots\,\}$, or the red and black membership relations $\color{red}{\in}$ and $\in$, are told apart at a glance.

5.1 Atoms out of the empty set

Inside the black copy, build the von Neumann numerals from the black empty set,

$$0 \;=\; \emptyset, \qquad n+1 \;=\; n \cup \{\, n \,\},$$

each a black, hereditarily finite set obtained, ultimately, from $\emptyset$. By black extensionality (the equations of the formal theory above) these are pairwise distinct: $n \neq m$ whenever $n \neq m$. Put

$$\mathbb{A} \;:=\; \{\, n \;:\; n \in \omega \,\} \;\subseteq\; \mathrm{Set}[X].$$

Two observations make $\mathbb{A}$ behave *as a set of atoms* for the red theory.

- **Admissibility.** By the second recursive knot $X = \mathrm{Set}[X]$, every black set is an admissible red atom; in particular each n is one.
- **Opacity.** The red membership relation $\in$ cannot see into a black set (the *Intuitions* above). Hence, *as far as the red theory can tell*, an element n of $\mathbb{A}$ has no internal structure whatsoever: the only facts the red theory may assert about n and m are $n = m$ or $n \neq m$, and those are settled, out of sight, by black equality.

Thus $\mathbb{A}$ presents to the red theory precisely as FM's set of atoms does: a countable family of mutually distinct, structureless objects that may sit inside sets but admit no membership query of their own.

Remark 1 (Where the risk went). This is the sense, promised in the introduction, in which atoms cost no infinite risk. ZFA and FM *postulate* an infinite set of primitive atoms; the postulate is the risk. Here nothing is postulated beyond $\emptyset$. The atoms are black hereditarily finite sets, and the stream $S[X]$ of the formal theory hands them out lazily, $\mathrm{take}(\Sigma, n)$ at a time, so that any red set ever built mentions only finitely many of them. The supply is *potentially* infinite, never an actual completed infinity: the commitment is the black empty set together with the same successor step that underwrites ω, and not one primitive atom more.

5.2 The permutation action

The stream operation $\mathrm{swap}(S[X], \rho)$ of the formal theory applies a finite permutation $\rho = \{\, j_1/i_1 \;:\; \cdots \;:\; j_k/i_k \,\}$ to the ordering of atoms; these finite permutations generate the full group $\mathrm{Sym}(\mathbb{A})$. A permutation $\pi \in \mathrm{Sym}(\mathbb{A})$ acts on red sets by red-membership recursion exactly as in FM,

$$\pi \cdot a = \pi(a)\ (a \in \mathbb{A}), \qquad \pi \cdot S = \{\, \pi \cdot e \;:\; e \in S \,\}.$$

Because π moves only atoms and never the black interior of an atom, and because the red operations $\emptyset$, $\cup$, $\cap$, $\setminus$ and $\mathrm{embrace}_i$ are visibly equivariant (a renaming commutes with each), the action is by automorphisms of the red theory and respects the equations of the formal theory. We have, verbatim, the $\mathrm{Sym}(\mathbb{A})$-action that FM requires.

5.3 Finite support is automatic

Write R for the class of well-formed red sets, taken modulo red equality, all of whose atoms lie in $\mathbb{A}$. The finitary rules of the formal theory — embrace_i has finite arity, Atomicity adjoins one atom, and $\cup, \cap, \setminus$ and Comprehension over finite sets preserve finiteness — yield only *hereditarily finite* red sets. For $S \in \mathsf{R}$ let $\text{atoms}(S) \subseteq \mathbb{A}$ be the finite set of atoms occurring hereditarily in S.

Lemma 1 (Support). *For every $S \in \mathsf{R}$ the finite set $\text{atoms}(S)$ supports S: if $\pi \in \text{Sym}(\mathbb{A})$ fixes $\text{atoms}(S)$ pointwise then $\pi \cdot S = S$. Moreover $\text{atoms}(S)$ is the* least *support, so $\text{supp}(S) = \text{atoms}(S)$.*

Proof (Proof sketch). That $\text{atoms}(S)$ supports S is an induction on the rank of S: an atom $a \in \text{atoms}(S)$ is fixed by hypothesis, and for a set $S = \{ e_1, \ldots, e_m \}$ a permutation fixing $\text{atoms}(S) \supseteq \bigcup_i \text{atoms}(e_i)$ fixes each e_i by induction, hence fixes S. Leastness uses that $\mathbb{A}$ is infinite: if $a \in \text{atoms}(S)$ then, choosing a fresh $b \notin \text{atoms}(S)$, the transposition $(a\,b)$ fixes $\text{atoms}(S) \setminus \{a\}$ yet moves S, so a lies in every support. Hence no proper subset of $\text{atoms}(S)$ supports S.

Corollary 1. *Every $S \in \mathsf{R}$ is hereditarily finitely supported, with $\text{supp}(S) = \text{atoms}(S)$. Freshness is occurrence-failure, $a \# S \iff a \notin \text{atoms}(S)$, and since $\mathbb{A}$ is infinite the fresh quantifier $\mathsf{И}$ and its some/any law (Section 4) hold in R.*

The point worth stressing is that in FM finite support is an *axiom*, imposed by cutting the cumulative hierarchy down to $\mathcal{U}_{\mathsf{FM}}$. Here it is a *theorem*: the finitary red rules never build anything that is not finitely supported, so the cut is vacuous and nothing need be assumed.

Remark 2 (Two points of precision). Lemma 1 rests on two facts worth making explicit. First, the group in play is the *finitary* symmetric group of $\mathbb{A}$ — the permutations generated by the transpositions that swap realises — for which supports are closed under intersection and least supports therefore exist; on an arbitrary subgroup they need not. Second, the action permutes atoms while treating each as structureless: it is an automorphism of the *red* structure $(\mathsf{R}, \in)$ alone, and is not asked to respect the black membership $\in$ internal to an atom. The atoms are accordingly black-distinct but red-opaque, and the entire support theory lives at the red level.

5.4 The representation theorem

Let $\mathcal{M}$ be the structure with universe $\mathbb{A} \cup \mathsf{R}$, with the atoms $\mathbb{A}$ as urelements, with $\in^{\mathcal{M}}$ the red membership relation and $=^{\mathcal{M}}$ red equality.

Theorem 1 (Representation). *$\mathcal{M}$ is a model of hereditarily-finite Fraenkel–Mostowski set theory: the FM analogue of $\text{HF}(\mathbb{A})$. Concretely, interpreting the atoms of FM as $\mathbb{A}$, its sets as R, and its membership as red $\in$, the structure $\mathcal{M}$ validates extensionality, the empty set, pairing, (finite) union, (finite) separation and foundation in their ZFA form, carries the $\text{Sym}(\mathbb{A})$-action of Section 5.2, and satisfies the finite-support axiom. The fresh quantifier and the some/any theorem hold throughout.*

Proof (Proof sketch). The atoms are membersless: red $\in$ cannot enter a black set, so no $a \in \mathbb{A}$ has red elements, and atoms are $\in$-minimal and disjoint from R in sort, as ZFA demands. Foundation holds because each red set is built at a finite rank over the atoms, so $\in^{\mathcal{M}}$ is well-founded. The remaining axioms come straight from the rules of the formal theory: Foundation gives $\emptyset$; embrace with Union gives pairing $\{ e_1 \} \cup \{ e_2 \}$ and, iterated, every finite set; Union gives binary and hence finite unions; Subtraction and Comprehension over finite domains give finite separation; and extensionality holds for the model provided provable red equality coincides with sameness of members — a property the equations of the formal theory are intended to force, and which Section 8 records as one of the obligations a fully rigorous development must discharge. The $\mathrm{Sym}(\mathbb{A})$-action and the finite-support axiom are Section 5.2 and Corollary 1; the some/any law follows because $\mathbb{A}$ is infinite and every $S \in \mathsf{R}$ has finite support. Powerset and infinity are absent — as expected of an HF universe — and are the subject of Section 5.6.

Corollary 2 (Two interlocked models). *By the red/black symmetry of the construction, the same argument run with the colours exchanged produces a black model $\mathcal{M}$ of hereditarily-finite* FM *whose atoms are the red numerals $\{ n : n \in \omega \} \subseteq \mathrm{Set}[X]$. The two models interlock: the atoms of each are sets of the other. In the game-semantic reading of the* Intuitions *above, player's atoms are opponent's strategies and conversely — a single reflective universe carrying both* FM *models at once, each underwriting the other's names.*

Remark 3 (What the HF *restriction costs).* It should be said plainly what is and is not delivered. The phenomena that give FM its character — finite support as a genuine constraint on *infinite* objects, equivariance with non-trivial orbits, and the some/any law quantifying over infinitely many fresh names against a fixed infinite support — all require infinite, finitely-supported sets, of which $\mathcal{M}$ has none. On the hereditarily finite fragment these properties hold, but trivially: every set is finite, so finite support is automatic and the some/any law degenerates. $\mathcal{M}$ is thus best described as a model of the *nominal hereditarily finite sets*: faithful to the ZFA and FM axioms, but exercising the distinctively nominal content only once the infinitary extension of Section 5.6 is in force. We retain the phrase "model of FM" only under this qualification.

5.5 Finitary by design: the hereditarily finite core

It is worth being explicit about why Theorem 1 lands on the hereditarily finite fragment, because the reason is also the source of the construction's advantage. The signature of the formal theory offers, as set constructors, only $\emptyset$, the finite-arity embraces $\mathrm{embrace}_i$ ($|\mathrm{embrace}_i| = i < \omega$), the binary $\cup, \cap, \setminus$, and Comprehension over finite domains $S_1 \times \cdots \times S_n$. Each of these maps finite sets to finite sets, and the only base case is $\emptyset$; so a trivial induction shows that *every* derivable red set is hereditarily finite. The model is $\mathrm{HF}(\mathbb{A})$, the nominal hereditarily finite sets over the atoms.

Two of FM's defining commitments are, in this regime, theorems rather than axioms.

- **Foundation** holds because every red set sits at a finite rank over the atoms, so $\in^{\mathcal{M}}$ is well-founded (Theorem 1); we never assumed regularity.

– **Finite support** holds because a hereditarily finite set mentions only finitely many atoms, each of which is its own least support (Lemma 1); we never assumed the finite-support axiom.

Correspondingly there is no infinity axiom and no infinite risk: the stream $S[X]$ is read only through $\mathrm{take}(\Sigma, n)$ for finite n, so at no point is an actually-infinite set formed. This is the regime in which the reflective theory strictly improves on axiomatic FM — both of the moves that define FM, foundation and the finite-support cut, are *derived* from the construction rather than *posited*.

5.6 Going infinitary, and the return of the cut

Full FM also has infinite, finitely-supported sets — $\mathbb{A}$ itself, or the equivariant powerset $\mathcal{P}_{\mathrm{fs}}(\mathbb{A})$ of finitely-supported sets of atoms — which the finitary signature cannot build. To reach them one adds an *infinitary* constructor. The most economical is an ω-ary (or κ-ary) embrace that consumes a whole stream rather than a finite prefix,

$$\frac{\Sigma; \Gamma \vdash S \qquad (\sigma, \Sigma') = \mathrm{take}(\Sigma, \omega)}{\Sigma';\, \sigma, \Gamma \vdash \mathrm{embrace}_\omega(\sigma) \cup S}\ \textsc{Infinitary-Embrace},$$

where $\mathrm{take}(\Sigma, \omega)$ returns the entire stream; equivalently one adjoins a powerset operator and an infinity axiom and climbs the red cumulative hierarchy $\mathcal{U}_\alpha(\mathbb{A})$ through all ordinals. Either way two things change, and — this is the point the finitary core conceals — they change *together*.

An actual infinity, hence one ω of risk. The lazy stream becomes a completed set; $\mathbb{A}$ itself becomes a red set. By Remark 1 the commitment incurred is exactly black infinity, a single ω-style axiom. No primitive atoms are reintroduced — the atoms remain black hereditarily finite sets built from $\emptyset$ — but the potential infinity of the finitary core is now actual.

Finite support is no longer free; the cut returns. Below ω, finite support was automatic. The infinitary hierarchy is not so forgiving: it contains sets that have *no* finite support. The canonical witness is the graph of the stream enumeration,

$$E \;=\; \{\, (n, a_n) : n \in \omega \,\}, \qquad a_n = \text{the } n\text{-th atom of } S[X],$$

which is fixed by a permutation π only when π preserves the enumeration, i.e. only when π is the identity on the infinitely many a_n; so no finite set of atoms supports E. This is no accident. The formal theory already tells us that $S[X]$ "may be thought of as an ordering on $\mathbb{A}$", and a total order on an infinite set of atoms is the textbook example of non-finitely-supported data. The finitary rules never expose that order — take reads only a finite prefix — which is precisely why support comes for free below ω. An infinitary embrace that swallows the whole stream imports the order, and with it the non-finitely-supported sets.

Consequently, in the infinitary regime one must do exactly what axiomatic FM does: cut down to the hereditarily finitely-supported part,

$$\mathcal{U}_{\mathsf{FM}} \;=\; \{\, x : x \text{ is hereditarily finitely supported}\,\},$$

discarding E and its kind. On the cut, the some/any theorem and the equivariance principle hold verbatim, and the result satisfies the Gabbay–Pitts axioms. The instructive point is that the cut is now the *only* thing borrowed from axiomatic FM: the atoms still come from the dual copy, the $\mathrm{Sym}(\mathbb{A})$-action still comes from swap, and foundation still comes from the construction (Section 7); only the finite-support axiom must be re-imposed, and only because we elected to admit actual infinities.

This locates precisely where reflection saves risk and where it cannot. Atoms: always free, from the dual copy. Foundation: always free, from local well-foundedness (Section 7). Finite support: free in the finitary core, an imposed cut once infinities are admitted. The finitary/infinitary boundary is therefore not a technicality but the exact ledger line between what the construction *derives* and what it must still *assume* — and a deliberate design choice for any theory built this way. We would keep the finitary signature as the canonical core, where the accounting is cleanest, and treat Infinitary-Embrace as an optional, clearly-priced extension.

6 Applications: nominal techniques founded on reflection

The original motivation for FM in computer science was Pitts and Gabbay's *nominal* treatment of names and binding [7]. That treatment is founded squarely on FM: names are the atoms, α-equivalence is an orbit relation under the permutation action, and the freshness machinery is exactly support and the И-quantifier. Since Section 5 realises FM inside the reflective universe — the nominal hereditarily finite fragment outright, and the full theory under the infinitary extension of Section 5.6 — the entire nominal edifice descends with it. The names that nominal techniques take as primitive become, here, black sets built from the empty set; the reflective construction *founds* the nominal theory rather than presupposing its atoms.

6.1 Nominal sets, internally

A *nominal set* is a set carrying a $\mathrm{Sym}(\mathbb{A})$-action in which every element is finitely supported; nominal sets and equivariant maps form the category **Nom**, equivalent to the Schanuel topos. The model $\mathcal{M}$ of Theorem 1 is, on the nose, a nominal set: the action is that of Section 5.2, and finite support is Corollary 1. Every construction nominal practitioners rely on — the equivariance principle (anything definable from equivariant data is equivariant), least supports, and the some/any law for И — is therefore available inside the reflective universe, with the colour discipline of Section 5 keeping the bookkeeping honest: names live in the black copy, the sets that bind them in the red copy.

6.2 Abstract syntax with binding

The headline application of [7] is a uniform account of inductively-defined syntax *modulo* α-equivalence. Its key device is *name abstraction*: for a nominal set X,

$$[\mathbb{A}]X \;=\; (\mathbb{A} \times X)\big/\!\sim, \qquad \langle a\rangle x \sim \langle b\rangle y \iff \text{И}c.\ (a\,c)\cdot x = (b\,c)\cdot y,$$

where $\langle a\rangle x$ is the class of (a, x) and $(a\,c)$ is a transposition. Both the action and И are present in $\mathcal{M}$, so $[\mathbb{A}]X$ is definable there, and it has the property that makes it a model of binding: $\mathrm{supp}(\langle a\rangle x) = \mathrm{supp}(x) \setminus \{a\}$, i.e. abstraction *discharges* the bound name from the support, precisely as a binder removes a variable from the free-variable set.

Inductive syntax is then obtained as usual. Object-level λ-terms, for instance, are the least solution of

$$\Lambda \;\cong\; \mathbb{A} \;+\; \Lambda \times \Lambda \;+\; [\mathbb{A}]\Lambda \qquad \text{(variable, application, abstraction)},$$

with α-equivalence built into the third summand. Structural recursion and induction come with the *freshness condition for binders*, justified by the some/any law: a definition or proof that treats a bound name as fresh for its parameters is sound because, $\mathbb{A}$ being infinite, "some fresh name" and "every fresh name" agree (Corollary 1). This is exactly the reasoning principle nominal users invoke when they "choose a fresh variable".

The risk accounting carries through and sharpens the point. Each individual term — each element of Λ — is a hereditarily finite nominal set, so it already lives in the HF model $\mathcal{M}$ of Theorem 1, founded on the empty set alone: a binding tree of any fixed shape costs no infinite risk. The completed datatype Λ, an infinite equivariant set, is one of the finitely-supported infinities of Section 5.6, and so costs exactly one ω — black infinity — and still not a single primitive name. Where Pitts and Gabbay begin with an infinite set of atoms given by fiat, here the names are manufactured, the α-structure is an orbit of the swap action, and the whole apparatus is refinanced down to one empty set and one successor.

6.3 Nominal algebraic theories

Beyond syntax, equational reasoning in the presence of names — nominal equational logic and the nominal Lawvere theories of Clouston [6] — takes its semantics in **Nom**. As $\mathcal{M}$ furnishes **Nom**-structure internally, any such theory (name-restriction, scope extrusion, binding signatures and their equational laws) can be interpreted in the reflective universe. The reflective setting adds one structural feature these accounts do not have: by Corollary 2 there are *two* interlocking name-universes, the red and the black, each supplying the other's names. Reading red as player and black as opponent recovers the game-semantic picture of the *Intuitions*: a name on one side is a strategy on the other, and a nominal theory and its dual coexist in a single reflective universe.

6.4 The hereditarily finite fragment is the right fragment

Remark 3 and Section 5.6 flagged the hereditarily finite restriction as the limit of what the finitary theory captures relative to full FM. From the standpoint of *computing*, that restriction is not a limitation but an alignment with practice. The realised nominal tools — FreshML and Fresh O'Caml [12,11], αProlog [5] — manipulate only hereditarily finite data: abstract syntax trees over a name supply, with name-swapping and nominal unification as the engine [10]. The completed set of atoms, the equivariant infinite sets, the Schanuel topos — these inhabit the *semantics* of those languages, never a running program. A program holds a finite term mentioning finitely many names and asks only local questions of it: are these two names equal, is this one fresh, swap that pair.

So the fragment on which the reflective construction is sharpest — where finite support is a theorem rather than an imposed cut, the some/any law holds, and no infinity and no infinite risk are incurred (Section 5.5) — is exactly the fragment these implementations inhabit. The hereditarily finite cut is where the theory and the practice coincide. This suggests a methodological point worth making explicit: a foundation for nominal *computing*, as distinct from nominal *set theory*, has no need to reach for infinite finitely-supported sets at all. It needs the nominal hereditarily finite sets, and those the reflective construction delivers from the empty set with the strongest guarantees and the least risk. What looked like the construction's ceiling is, for the applications that motivated the whole enterprise, precisely its home.

6.5 Reflective names: a more expressive theory of names

There is a second, less obvious dividend, and it concerns the *nature of names* rather than their supply. In the Pitts–Gabbay world an atom is primitive and structureless: the only operations on it are equality, freshness, and swapping, and that opacity is the entire point. It is what forces every definable function to be equivariant, and equivariance is what makes α-equivalence and capture-avoidance come for free [7]. In the reflective construction a name is *not* primitive: it is a black set, and one may cross the colour barrier to look inside it.

This is the set-theoretic shadow of the move the reflective higher-order calculus makes against the π-calculus [9]: where π names are atomic, reflective names are quoted processes, and reflection lets one pass between a name and the structure it names. The reflective set theory says that "names as quoted structure" is not special to process calculi — it is a general operation on theories, two copies each serving as the other's atoms — and that performing it on set theory yields nominal sets whose names carry structure. What such names buy, that opaque oracle-drawn names cannot, is fourfold, and it tells in the distributed and mobile setting rather than in pure syntax.

- **Names are computed, not drawn.** "Fresh" becomes "a black set not occurring in this finite red structure" — decidable and constructive, a deterministic fresh-name *function* rather than the observable side effect that name generation is in FreshML. No gensym oracle and no central allocation authority are required, so in a decentralised system any party may mint and verify a name locally with no coordination.

– **Names are first-class data.** A name is a finite set, so it serialises, communicates, and hash-conses; content-addressed names — names that *are* the canonical encoding of what they name — fall straight out.
– **Names have a decidable structural identity**, rather than an abstract primitive equality.
– **Self-representation.** The two copies form a reflection tower: the metalanguage's names are the object theory's sets and conversely, which is a principled basis for metaprogramming and staging that a flat primitive name type cannot express.

The honest cost fixes the shape of the contribution. Reflection breaks opacity, and opacity was load-bearing: a program that inspects a name's black structure is not equivariant, so it need not respect α-equivalence, and for it the automatic capture-avoidance and the some/any principle are forfeit. The value is therefore not to discard opacity but to obtain a *unifying* metalanguage with a knob. Staying within one colour is ordinary nominal programming, with every guarantee intact, recovered as the *equivariant fragment*; crossing colours is the additional reflective power. A metalanguage built on this would carry a grading — a modality distinguishing equivariant, colour-respecting computations from reflective ones — so that its type system states exactly which operations still enjoy the nominal guarantees and which have traded them for structure. That discipline, in the spirit of graded and modal type systems, is the genuine open problem here: we can claim the unification and the semantic account, but the better metalanguage — a reflective FreshML in which the Pitts–Gabbay discipline is the equivariant sublanguage and reflective names are the whole — is the research programme this construction opens, not an artifact it delivers.

The chain is short and worth stating plainly. Nominal techniques rest on FM; FM embeds in the reflective universe (Section 5); the reflective universe is built from the empty set. Composing, the Pitts–Gabbay theory of names and binding is founded on two recursively intertwined copies of ordinary set theory, with its primitive atoms replaced by constructed black sets and its infinite risk replaced by the finite risk of $\varnothing$ — exactly the reconciliation announced in the introduction, now delivered for the application that made FM matter to computer science in the first place. Two further observations sharpen the point for computing rather than for set theory. The hereditarily finite cut, a ceiling for the set theory, is the natural home of the realised nominal languages (Section 6.4); and reflective names, by trading opacity for structure under a graded discipline, promise a metalanguage that contains the nominal one as its equivariant fragment while reaching the computed, serialisable, content-addressable names of the mobile setting (Section 6.5).

7 A bridge between well-founded and non-well-founded set theory

ZF and FM are well-founded theories: both take the axiom of foundation, so membership admits no infinite descent and no cycles. At the opposite pole stands Aczel's anti-foundation axiom (AFA), which replaces foundation outright, admits sets such as

$\Omega = \{\Omega\}$, and gives every accessible pointed graph a unique decoration as a set [1,3]. The reflective construction sits between these poles, and clarifies their relation, because here non-well-foundedness is not *posited* but *produced*: it is what two well-founded theories generate when they are intertwined.

7.1 Each colour is locally well-founded

Restricted to a single colour, the construction is as well-founded as ZF. The red membership $\in$ relates red sets only to their red members, bottoming out at $\emptyset$ and at the red atoms; and every red set has a rank over those atoms (finite in the core of Section 5.5, an ordinal in the infinitary extension of Section 5.6). So $\in$ is well-founded, and foundation holds for the red theory — as a theorem, never assumed (Theorem 1). The black theory is well-founded for the identical reason. Crucially, an atom is *opaque* to the membership that has it as an atom: $\in$ cannot descend into a black atom, and $\in$ cannot descend into a red one. Each colour's well-foundedness rests exactly on this opacity.

7.2 The pair is globally non-well-founded

Now erase the colour barrier. Let $\in_\star$ be the *colour-blind* membership that refuses to treat atoms as opaque: where a red set has a black atom $a = b$, the relation $\in_\star$ exposes the black members of b, and symmetrically. Under $\in_\star$ a red set may contain a black set, which contains a red set, which contains a black set, without end:

$$S_0 \ni_\star b_0 \ni_\star S_1 \ni_\star b_1 \ni_\star S_2 \ni_\star \cdots,$$

an infinite descending $\in_\star$-chain that alternates colours. Globally, foundation fails. And the failure is not marginal: the two universes are themselves *defined* by the mutually recursive equations

$$X = \mathrm{Set}[X], \qquad X = \mathrm{Set}[X],$$

a system whose solution is a fixed point. Read as a specification of objects by their unfoldings, this is precisely the kind of non-well-founded equation that Aczel's solution lemma is designed to solve [1]; allowing a red name whose black unfolding refers back to it realises genuine Ω-like reflexive structure under $\in_\star$.

7.3 Non-well-foundedness as an interaction effect

This is the conceptual dividend. In Aczel's theory non-well-foundedness is an *axiom*: AFA is asserted in place of foundation, of a single universe. In the reflective construction it is an *emergent global property*: well-foundedness is a feature of each colour *in isolation*, non-well-foundedness a feature of the colours' *interaction*. The colour barrier is the exact seam between the two — intra-colour $\in$ is well-founded, inter-colour $\in_\star$ is not — so one can read off where descent terminates and where it loops simply by tracking colour changes.

The coalgebraic gloss makes the spectrum sharp. A mono-chromatic theory is an *initial algebra* of its set-builder operator: the least fixed point, well founded, the home of ZF and

FM, and the object of study of *algebraic set theory*, where the well-founded universe appears as the free algebra for a set-builder operator on a category of classes [8,2]. The intertwined system (Set[−], Set[−]), taken instead as a *greatest* fixed point, is a *final coalgebra*: non-well-founded, the home of AFA [1,3]. A caveat of size belongs here, and it falls exactly along the finitary/infinitary seam of Sections 5.5–5.6. The *finite* powerset functor — the operator behind the finitary core — is finitary, so its final coalgebra exists already at the level of sets, and the non-well-founded reading of the HF theory is literal. The *full* powerset operator has no final coalgebra at all, by Cantor's theorem together with Lambek's lemma; the unrestricted, infinitary theory therefore lives not in a set-sized final coalgebra but in Aczel's class-sized framework of accessible pointed graphs and the solution lemma [1]. The reflective construction places both fixed points of one system of operators within a single framework, and the knobs of this paper — least versus greatest fixed point, finitary versus infinitary embrace (Section 5.6), acyclic versus cyclic colour discipline (Section 9) — are exactly the controls that slide a theory along the spectrum from ZF and FM at the well-founded end to Aczel's universe at the non-well-founded end. Making that correspondence precise — decorating every accessible pointed graph by a colour-blind set picture, and exhibiting the finitary intertwined universe as a genuine final coalgebra — is a natural next step (Sections 8 and 9). What the present construction already shows is the structural moral: *non-well-foundedness need not be assumed; it can be manufactured, as a global property of locally well-founded theories.*

8 Toward a rigorous construction

The development to this point is programmatic — the constructions above are pitched at the level of intent: they say what the theory is meant to be and exhibit the model it is meant to carry, but several of the moves are made by notation where they ought to be made by theorem. We close by setting down, as a checklist of definitions to fix and obligations to discharge, what a fully rigorous development would require. We group them under the headings a careful referee would use — the fixpoint, the metatheory, the membership theory, comprehension, and the construction's place among its neighbours — and note that for the finitary core each obligation reduces to a statement about hereditarily finite sets.

8.1 The knotted fixpoint, via algebraic set theory

The central move, $X = \mathrm{Set}[X]$ and $X = \mathrm{Set}[X]$, is at present a definition by notation; it must be exhibited as a fixed point that provably exists. Algebraic set theory (AST) is, we think, the right basis for this, because it is built precisely to produce universes of sets as fixed points of a power*class* operator while dodging the size obstruction that defeats the naive power*set* construction [8,2].

Recall its shape. One works in a Heyting category $\mathcal{C}$ of *classes* with a distinguished collection of *small* maps and a *powerclass* functor $\mathcal{P}_s\colon \mathcal{C} \to \mathcal{C}$, sending a class to the class of its *small* sub-objects. The universe of sets is the free $\mathcal{P}_s$-algebra: an object U

with a structural isomorphism $\mathcal{P}_s U \cong U$, and the central theorem is that this initial ZF-algebra exists in any category with small maps and models the axioms of set theory, with membership recovered from the algebra structure. Restricting $\mathcal{P}_s$ to *small* sub-objects, rather than arbitrary sub-classes, is exactly what makes $\mathcal{P}_s U \cong U$ possible where Cantor forbids $\mathcal{P}X \cong X$ — so the size problem flagged in Section 7 is dissolved at the level of the framework, not patched case by case.

Two features of AST fit the present construction almost too well. First, it handles atoms by a coproduct: the cumulative hierarchy over an object A of atoms is the initial algebra of $Y \mapsto A + \mathcal{P}_s Y$, in which the points of A acquire no members — they are atoms because the free construction *forgets* whatever structure they carried. That forgetfulness is precisely our colour opacity (Remark 2): red sees the points of its atom-supply but not their black interior. Second, the universe is constructed as a W-type quotiented by a bounded bisimulation, so the development is constructive and portable — and, not incidentally, close to what a formalisation would have to build.

The knot is then a two-sorted instance. Take the endofunctor on $\mathcal{C} \times \mathcal{C}$

$$H(R, B) \;=\; \big(\, B + \mathcal{P}_s R, \;\; R + \mathcal{P}_s B \,\big),$$

whose initial algebra is a pair $({\color{red}X}, X)$ with ${\color{red}X} \cong X + \mathcal{P}_s {\color{red}X}$ and $X \cong {\color{red}X} + \mathcal{P}_s X$: each colour's universe is its own sets together with the other colour's universe *as atoms* — precisely ${\color{red}X} = \mathrm{Set}[X]$ and $X = {\color{red}\mathrm{Set}[X]}$. The obligation thus becomes to run the AST existence and modelling theorems for H in place of $\mathcal{P}_s$ alone. The ingredients are unchanged — coproduct, powerclass, W-types, bounded bisimulation — so one expects the same proof to go through two-sorted; what it would establish is that the knotted universe *exists* and that each colour *models* ZFA, discharging in a single stroke both this obligation and the extensionality and separation obligations of Section 8.3.

Three riders complete the picture. For the *finitary* core one replaces $\mathcal{P}_s$ by the finite-powerclass functor; being finitary, the initial algebra of the corresponding H exists already at the level of *sets*, as the colimit of the initial ω-chain $\varnothing \to H(\varnothing) \to H^2(\varnothing) \to \cdots$, with no class machinery required — and it is the HF universe of Section 5. For the non-well-founded reading of Section 7 one takes the *final coalgebra* of H in place of the initial algebra: at set size for the finitary functor, and in the class framework — where it coincides with Aczel's universe — for the full one. And a caveat worth stating: the initial ZF-algebra models *constructive* set theory by default, so recovering classical ZFA/FM requires the corresponding classical principles (power set, full separation, excluded middle) on the class category. With those in place, AST supplies exactly the existence theorem we have so far only gestured at, and supplies it in a form that already carries the membership and extensionality structure the rest of this section asks for.

8.2 Metatheory and consistency strength

We nowhere name the theory in which the construction is conducted, and the headline claim — atoms without infinite risk — has content only relative to a fixed metatheory. The obligation is to state that metatheory and to cash the risk claim as a statement of interpretability. The natural target is hereditarily finite set theory, equivalently first-order arithmetic: the model $\mathcal{M}$ of Theorem 1 is itself an interpretation of the finitary reflective theory in HF, so the two are equiconsistent and the "risk" is exactly that of arithmetic — no infinite set of primitive atoms is ever posited. The representation theorem, in other words, already *is* a relative-consistency proof; what remains is to state it as one. The infinitary extension of Section 5.6 raises the target by a single ω: it is interpretable in Z — or in ZF with foundation reinstated per colour — the lone infinity being black infinity. Making "finite versus infinite risk" precise in this way is, we expect, the first thing a set theorist would ask for, and it is the claim on which the paper's title rests.

8.3 Extensionality and the membership theory

Extensionality is not visibly secured by the equations of the formal theory, which give reflexivity, symmetry, transitivity, and idempotence of $\cup$ and $\cap$, but not that sets with the same members are equal. A rigorous development must either adopt extensionality as an axiom or derive it, and must then prove that provable red equality coincides with sameness of members — the property the proof of Theorem 1 leans on. The membership rules need a matching completeness theorem. As presented they decide $\in$ on unions, intersections, differences, and singletons, but no rule governs membership in a nested $\{\, S \,\}$, and one wants a proof that for HF sets every true $e \in S$ and $e \notin S$ is derivable and that the relation is decidable. Until then, the remark that the syntactic theory is "too fine grained" and is corrected by equations is a programme rather than a theorem. We note that the algebraic-set-theory route of Section 8.1 would discharge this obligation along with existence: its modelling theorem delivers extensionality for the constructed universe directly, so on that route the equational theory need only be shown *sound* for the model rather than independently complete.

8.4 Comprehension and freedom from paradox

The Comprehension rule forms $\{\, f(s_1, \ldots, s_n) : s_i \in S_i \,\}$ for a function $f\colon S_1 \times \cdots \times S_n \to S$, and the admissible class of f is left open although it governs both the strength of the theory and its safety. The rule is the image of a function over sets already built — separation- and replacement-flavoured rather than naive comprehension — so the expected outcome is that no Russell set arises; but this must be argued, with care in the cross-colour case, where f must respect the opacity of atoms. The cleanest argument is semantic and is already to hand: the HF model witnesses consistency, so fixing f to range over the metatheory's set functions and reading the rule in $\mathcal{M}$ shows the theory paradox-free relative to HF. Specifying f precisely, and confirming that the cross-colour instances preserve well-formedness, is the remaining work.

8.5 Placing the construction

Finally, the construction should be located among its neighbours, which would both guard against reinvention and sharpen its claims. Three comparisons are immediate. The permutation action and the atoms align it with the Fraenkel–Mostowski permutation models and their nominal descendants [7]; the non-well-founded reading aligns it with Aczel [1]; and the colour barrier — red cannot see into black — is a two-level *stratification*, inviting comparison with the typed and stratified set theories. Most pointedly, red and black *mutually interpret* one another, so the construction is an instance of mutual interpretation between set theories; whether the two copies are *bi-interpretable*, and in what precise sense the universe of set theories promised by the abstract is a *multiverse*, are exactly the questions that would connect this work to the contemporary study of interpretations between set theories. We regard settling them as the natural completion of the present work.

None of these obligations looks out of reach for the finitary core, where each reduces to a statement about hereditarily finite sets and several are corollaries of the model already built. Together they are what would carry the construction from a suggestive picture to a theorem — and, not incidentally, they are close to the list of lemmas a formalisation would have to state in any case.

9 Conclusions and future work

We set out to reconcile two things usually held apart: a set theory with atoms, of the kind nominal techniques require, and the austerity of building everything from the empty set. The reconciliation is the knot. Taking two copies of set theory — a red and a black — and tying them so that each colour's atoms are the other colour's sets, we obtain an infinite supply of atoms without positing a single one: every atom is a set of the dual colour, manufactured from $\varnothing$, and the infinite risk that Chaitin's accounting charges to a primitive infinity of atoms is refinanced as the ordinary risk of one empty set and one successor.

Around this knot we have assembled a connected picture. The formal theory above fixes the two copies and ties them; a self-contained sketch of Fraenkel–Mostowski set theory (Section 4) makes the target precise; and the representation theorem (Section 5) exhibits the red hereditarily finite sets over the black-set atoms as a model of the nominal hereditarily finite sets, in which finite support — imposed as an axiom in FM — is instead a theorem. The boundary between the finitary core and its infinitary extension proved to be the exact ledger line between what the construction *derives* — atoms, foundation, finite support — and what it must still *assume*: a single ω, and the finite-support cut, once actual infinities are admitted.

The construction earns its keep in three further ways. It founds the Pitts–Gabbay nominal techniques (Section 6), and on exactly the hereditarily finite fragment that the realised nominal languages inhabit, so the restriction that bounds the set theory is the natural home of the computing; it suggests, through reflective names, a more expressive theory of names in which the nominal discipline reappears as an equivariant fragment; and it serves as a bridge (Section 7) between the well-founded theories ZF and FM and Aczel's non-well-founded universe, by making non-well-foundedness not an axiom but an emergent, global property of two locally well-founded theories. The road from this programmatic account to a rigorous one runs, we have argued, through algebraic set theory (Section 8), whose initial-ZF-algebra theorem is purpose-built to supply the knotted fixpoint and, with it, the membership and extensionality structure the construction needs.

Several directions remain open; we record the ones we find most pressing.

9.1 Arbitrarily many colours

Two colours were a convenience, not a necessity. Replace the pair by any set C of *colours*; each $c \in C$ carries its own copy of the theory, $\mathrm{Set}_c[X_c]$, with its own membership $\in_c$ and its own comprehension $\{\cdots\}_c$. The only remaining datum is the *nesting discipline*: which colours may serve as atoms for which. This is a relation on C, and it is most naturally presented as a directed graph — equivalently a finite-state machine — with states C and an edge $c \to c'$ whenever c'-sets are admissible c-atoms. The atom supply of each colour is then the disjoint sum over its outgoing edges,

$$X_c \;=\; \bigsqcup_{c \to c'} \mathrm{Set}_{c'}[X_{c'}],$$

and the two-colour theory of this paper is the instance given by the two-state automaton red $\rightleftarrows$ black, with $X = \mathrm{Set}[X]$ and $X = \mathrm{Set}[X]$.

Everything in the paper relativises to this picture. The well-formed cross-colour nestings are the runs of the automaton; the local/global well-foundedness analysis of Section 7 reads off its structure, an acyclic colour graph giving a globally well-founded universe and a cycle giving non-well-foundedness along that cycle. (A self-loop $c \to c$ would let c-sets be their own atoms and collapse the barrier; the two-cycle red $\rightleftarrows$ black is the smallest discipline that keeps every colour locally well-founded while making the system globally non-well-founded.) This generalization is routine, and we record it mainly to fix the slogan: *a reflective set theory is an automaton on colours*, and the construction above is its two-state instance.

9.2 Generating colours dynamically

The genuinely interesting question we leave open. A fixed automaton fixes the palette in advance. One would instead like colours to be *generated on demand*: a fresh colour spawned in the course of building a set, the palette growing as construction proceeds, the automaton unfolding rather than given. The difficulty is not in admitting infinitely many colours — a countable palette is unremarkable — but in doing so *elegantly*, with a generation discipline internal to the theory, so that the mechanism producing colours is of a piece with the mechanism producing sets and names.

There is a suggestive circularity to exploit, which is why we think the question worth posing rather than dismissing. Generating a fresh colour is generating a fresh name, in exactly the sense that Sections 4–6 reconstruct: a colour is drawn from a supply, distinct from those already in play, and otherwise structureless. So a reflective theory might be made to *generate its own colours* by the very nominal machinery it models — the palette drawn, like the atoms, from a stream, and the colour automaton itself becoming a nominal object in a theory one level up. Whether this reflexive step can be taken without infinite regress, what its fixed point looks like, and whether it lands back inside the spectrum of Section 7, are the questions we find most worth pursuing. We do not settle them here; we mark them as the natural sequel to this work.

9.3 A semantics for the rho calculus

We close with a conjecture rather than a construction. Aczel gave a semantics for Milner's CCS inside non-well-founded set theory [1]: a process is identified with the set of its labelled transitions, $\{\,(a, P') : P \xrightarrow{a} P'\,\}$, so that a looping process is a non-well-founded set, bisimilar processes are equal, and the anti-foundation axiom's solution lemma supplies and individuates processes as the unique solutions of their behavioural equations. We surmise that the knotted universe developed here is the natural home for the same construction carried out for the reflective higher-order calculus, the rho calculus [9].

The fit is suggestive on two counts. First, the non-well-foundedness that Aczel imports by axiom is here available for free, as the global, colour-blind structure of the knotted universe (Section 7); the recursion that process behaviour requires is the recursion the knot already carries. Second — and this is what sets rho apart from CCS and the π-calculus — rho's names are not primitive but *reflected*: the quote operator @ turns a process into a name, the dereference operator $*$ turns a name back into the process it quotes, and $*@P \equiv P$. In the knotted universe a name may be taken to be an *atom*, and atoms here are constructed sets of the dual colour, opaque until one crosses the colour barrier (Sections 5.1 and 6.5). Reading @ as *atom construction* — a process, presented as a set, is sealed into an atom of the opposite colour — and $*$ as *atom unpacking* — the barrier is crossed to recover the set the atom seals — the reflection law $*@P \equiv P$ becomes the round trip of these two operations.

We do not carry the construction out, and we do not claim its details are routine: communication, the substitution of names for names, and rho's equational theory would each have to be matched against their set-theoretic shadows, and the well-founded versus non-well-founded reading (Section 8.1) settled. We claim only that the ingredients line up — a calculus whose names are quoted processes asks for a set theory whose atoms are sets, and whose dynamics asks for non-well-foundedness, and the knotted universe offers both at once — and we conjecture that such a semantics exists and is more convenient than one built over an unstructured atom supply, precisely because the @/$*$ reflection it must model is already the construction's defining move.

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